XR Event Planning

Interactive Prototype 2b Testing Results

Course: DECO2300 Digital Prototyping and Extended Reality

1. Testing Overview

The final IP2b testing session evaluated whether participants could independently begin planning an event, adapt an existing furniture arrangement, and customize the venue using physical VR interactions.

All six participants, including five peers and one tutor, completed the tested tasks independently without facilitator assistance. The testing also collected feedback on the Welcome screen, starting furniture, spatial layout, interaction guidance, and physical controller usability.

2. Testing Results

The testing followed the four tasks established in the IP2b Testing Plan.

Scoring: Pass (2) = independent completion, Assist (1) = completion with facilitator assistance, Fail (0) = incomplete after assistance.

Interaction	P1	P2	P3	P4	P5	P6
Task 1: Begin planning	2	2	2	2	2	2
Task 2: Edit furniture layout	2	2	2	2	2	2
Task 3: Remove furniture	2	2	2	2	2	2
Task 4: Change theme	2	2	2	2	2	2
Task 4: Adjust lighting	2	2	2	2	2	2

The results table includes Task 1 (initial orientation) and an additional participant (P6). Guide and interaction hint use are documented separately in the following section.

All six participants independently completed Task 1 within the planned 30-second threshold. Exact completion times were not recorded, but I observed that each participant activated Start and initiated a relevant planning action within the required period.

All six participants also completed furniture arrangement, removal, lighting adjustment, and theme changes without assistance. However, some reported minor difficulties with furniture positioning, rotation, the lighting lever, and theme wand placement.

Guidance Use

- **P1:** Noticed Grab/Rotate hints and found them helpful. Did not use the Guide.
- **P2:** Noticed furniture interaction hints. Did not use the Guide.
- **P3:** P3: Used the Guide to check the controls and also found the interaction hints helpful, especially for rotation.
- **P4:** Noticed the Grab hint. Did not use the Guide.
- **P5:** Noticed furniture interaction instructions and found them useful. Guide use was not specifically documented.
- **P6:** Guide and hint use were not documented.

3. Participant Feedback Summary

Participant	Positive Feedback	Difficulty	Suggested Improvement
P1	Clear Start button, helpful starter furniture and hints	Chair rotation required several attempts	More furniture options
P2	Understandable menu and useful starting layout	Precise furniture positioning required adjustments	Furniture alignment aid
P3	Welcome screen and central carpet helped orientation	Lighting lever took time to get used to	More lighting options
P4	Starting layout made the venue feel more complete	Wand needed careful positioning in the holder	More decorations and lighting types
P5	Existing stage and tables helped communicate the layout	Initial controller grabbing felt awkward	Ability to save layouts
P6	Understood menu and completed all interactions	Menu too far right and floor-level furniture movement felt awkward	Better UI positioning and interaction comfort

4. Results Against Testing Assumptions

A1: First-Entry Orientation

All six participants successfully activated Start and initiated a relevant planning action independently within the planned 30-second threshold.

P1–P5 described the Welcome screen, Start button, or Event Planner menu as clear and easy to understand.

These results support A1, suggesting that the redesigned starting experience helped participants understand where to begin. Although all participants met the time threshold, individual completion times were not recorded, so differences in how quickly they began cannot be compared.

A2: Spatial Planning

All six participants independently completed the furniture editing task.

P1, P2, P4, and P5 described the starting arrangement as useful for understanding the space or deciding where to place additional furniture. P3 specifically identified the central carpet as a cue to keep the middle area clear.

These findings support A2, suggesting that the starting furniture and environmental details helped participants make layout decisions. However, individual furniture overlap and aisle-clearance observations were not documented separately, limiting further analysis of placement accuracy.

A3: Physical Customisation

All six participants independently completed furniture removal, lighting adjustment, and theme changes using physical VR interactions.

Despite successful completion, some participants experienced minor difficulties. P3 needed time to become familiar with the lighting lever, while P4 needed to position the theme wand carefully.

The tutor also identified awkward furniture movement because the controller had to be directed towards the floor.

These results support A3 in terms of independent completion, while identifying remaining opportunities to improve physical interaction comfort.

A4: Interaction Guidance

P1, P2, P3, P4, and P5 explicitly reported noticing furniture interaction hints. P1, P2, and P5 described the hints as helpful for understanding or remembering the controls.

P3 used the Event Planning Guide and reported that it helped.

These findings provide positive evidence for A4. However, Guide use was not documented for every participant, and most participants completed their interactions without opening it. Therefore, the results support the usefulness of the hints but provide more limited evidence about how effectively the Guide helps users recover from difficulties.

5. Testing Documentation Limitations

The testing results were based on recorded task outcomes, facilitator observations, and the supplied participant conversations.

Although all six participants independently completed the tasks, some details were not recorded individually:

- Exact time to Start activation and first planning action. All participants were observed meeting the 30-second threshold, but specific times were not retained.
- Detailed furniture overlap and aisle-clearance observations for each participant.
- Previous VR experience and recording details for each participant.
- Guide use for P5.

The participant conversations in Appendix A provide the supplied feedback from P1–P5. The tutor feedback in Appendix B was reconstructed from the facilitator's account and is not a verbatim transcript.

These limitations will be considered when interpreting the results in Design Evaluation 3.

Appendix A: Participant Post-Test Conversations

The following conversations reproduce the participant feedback supplied for the testing session.

Participant 1 (P1)

Facilitator: What helped or confused you when deciding where to begin?

P1: I think it was pretty clear. I saw the Start button first, and after pressing it I could see the furniture options, so I just went from there.

Facilitator: How did the starting furniture and room details affect your layout choices?

P1: It helped actually, because there were already tables and chairs. I didn't have to figure out the whole layout from scratch.

Facilitator: Which interaction was hardest, physically or otherwise, and why?

P1: Maybe rotating the chair. I had to try it a couple of times to get the position I wanted, but it was fine after that.

Facilitator: Did you notice or use the hints or Guide?

P1: Yeah, I saw the Grab and Rotate hints. They were pretty helpful, so I didn't really need to open the Guide.

Facilitator: What one change would most improve planning your event?

P1: Maybe having more furniture options, but overall I think it worked well.

Participant 2 (P2)

Facilitator: What helped or confused you when deciding where to begin?

P2: Nothing really confused me. I just pressed Start and looked at the menu. It was quite easy to understand.

Facilitator: How did the starting furniture and room details affect your layout choices?

P2: I liked that the venue already looked like an event space. It was easier to decide where to put the sofa and plant because I could see where everything else was.

Facilitator: Which interaction was hardest, physically or otherwise, and why?

P2: Probably moving the furniture to the exact spot I wanted. It wasn't really difficult, I just had to adjust it a bit.

Facilitator: Did you notice or use the hints or Guide?

P2: Yeah, I noticed the hints when I pointed at the furniture. They were clear enough, so I didn't use the Guide.

Facilitator: What one change would most improve planning your event?

P2: Maybe something that helps line up the furniture more neatly.

Participant 3 (P3)

Facilitator: What helped or confused you when deciding where to begin?

P3: The welcome screen helped. I knew where to start, and once the menu appeared it was easy to see what I could add.

Facilitator: How did the starting furniture and room details affect your layout choices?

P3: The middle carpet was actually helpful. I assumed that area should stay clear, so I placed the sofa and plant on the side.

Facilitator: Which interaction was hardest, physically or otherwise, and why?

P3: I think the lighting lever took a little getting used to. But once I moved it, I could see the brightness changing.

Facilitator: Did you notice or use the hints or Guide?

P3: Yeah, I checked the Guide just to make sure I understood the controls. Also the hints were helpful, especially for rotating things.

Facilitator: What one change would most improve planning your event?

P3: Maybe having more lighting options instead of just changing the brightness.

Participant 4 (P4)

Facilitator: What helped or confused you when deciding where to begin?

P4: It was straightforward. The Start button was obvious, and I could figure out the menu pretty quickly.

Facilitator: How did the starting furniture and room details affect your layout choices?

P4: I think it made the room feel more complete. I could imagine an actual event happening there, so I just made some small changes to the layout.

Facilitator: Which interaction was hardest, physically or otherwise, and why?

P4: Maybe putting the wand into the holder. I needed to position it properly, but I managed to do it.

Facilitator: Did you notice or use the hints or Guide?

P4: I noticed the Grab hint, yeah. I didn't really use the Guide because I could work out what to do.

Facilitator: What one change would most improve planning your event?

P4: I think adding more decorations would be nice, like flowers or different types of lights.

Participant 5 (P5)

Facilitator: What helped or confused you when deciding where to begin?

P5: I think everything was pretty easy to follow. I pressed Start, looked around, and then started adding furniture.

Facilitator: How did the starting furniture and room details affect your layout choices?

P5: It was useful because I could see the stage and tables already there. It gave me an idea of how the venue was supposed to look.

Facilitator: Which interaction was hardest, physically or otherwise, and why?

P5: Probably grabbing the furniture with the controller at first. But once I got used to it, it was okay.

Facilitator: Did you notice or use the hints or Guide?

P5: Yeah, I noticed the instructions when I pointed at the furniture. They made it easier to remember the controls.

Facilitator: What one change would most improve planning your event?

P5: Maybe being able to save the layout and come back to it later.

Appendix B: Tutor Feedback (P6)

Note: The following is feedback based on the tutor's testing session.

Testing Outcome

The tutor completed all tested tasks independently without facilitator assistance.

Furniture Menu

The tutor was able to see the furniture menu and understand the available options, including the buttons for adding tables and chairs. However, the menu was positioned too far to the right when viewed through the Meta Quest headset. Although the buttons remained accessible and understandable, the placement was not ideal for comfortable viewing.

Furniture Movement

The tutor also found furniture movement slightly awkward because the controller had to be directed towards the floor while repositioning objects. The prototype intentionally keeps furniture attached to floor level to prevent unrealistic placement. This feedback therefore raises a question about the comfort of the interaction rather than necessarily indicating that furniture should be freely lifted.

Overall Result

The tutor successfully completed all interactions without assistance. Their feedback identified opportunities to improve the comfort and accessibility of the interface while preserving the intended floor-constrained furniture movement. These are possible future improvements rather than changes that have already been implemented.